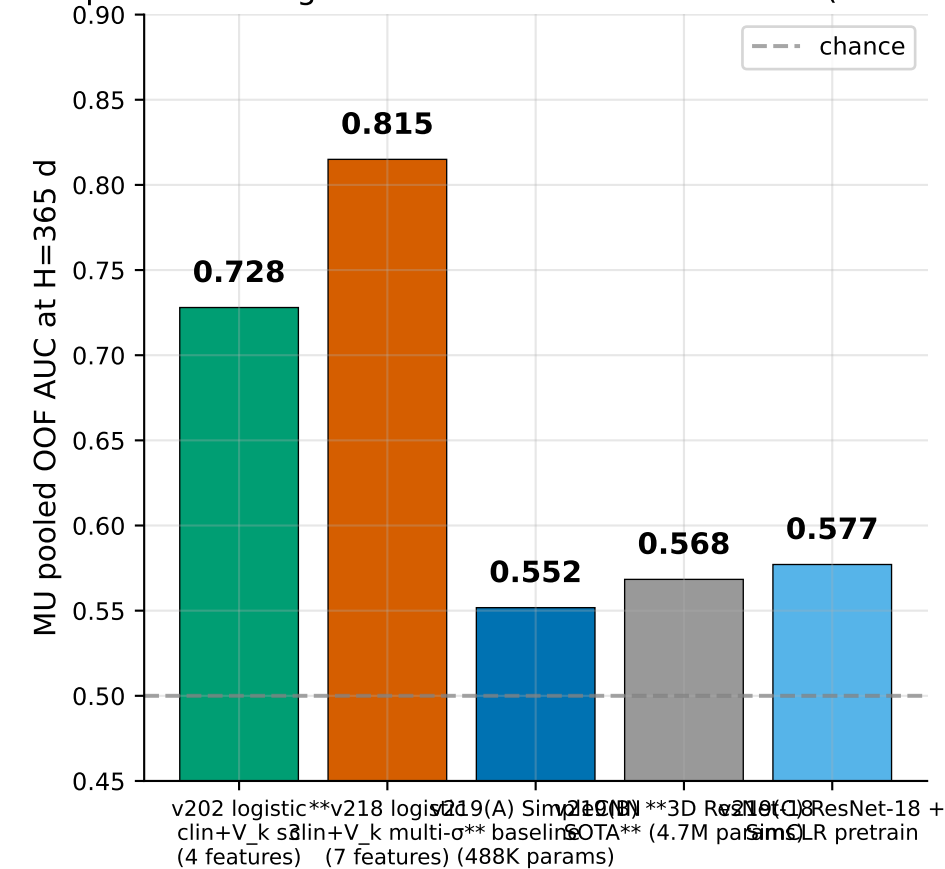
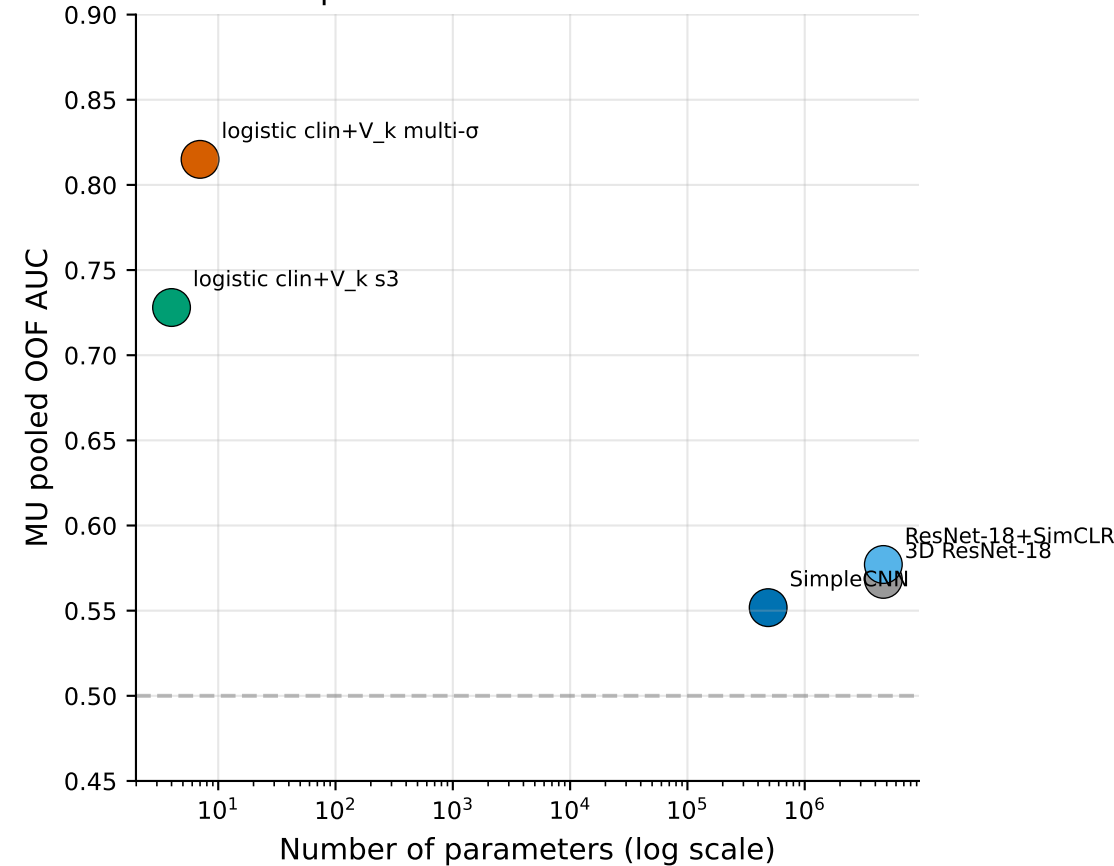


v219 BEYOND-NMI SOTA architecture comparison: 3D ResNet-18 (4.7M params, 9.6× SimpleCNN) achieves only AUC=0.568, FAILING to match the simple 7-feature logistic+multi- $\sigma$  V\_kernel (0.815). At MU n=130, more parameters HURT — the bimodal kernel is the optimal feature.

A. Logistic vs SOTA architectures  
Simple multi- $\sigma$  logistic CRUSHES SOTA ResNet-18 (+0.247 AUC)



B. Parameter count vs AUC  
More parameters HURT at MU n=130



C. Per-fold AUC distribution  
ResNet-18 high variance (overfit at n=130): range 0.51-0.99

